Kubernetes – Services

The Deployment.yaml

```
GNU nano 8.3                                   Deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80

                                   [ Read 19 lines ]
^G Help      ^O Write Out   ^F Where Is    ^K Cut       ^T Execute   ^C Location   M-U Undo      M-A Set Mark   M-] To Bracket  M-B Previous
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify   ^/ Go To Line M-E Redo      M-6 Copy       ^B Where Was    M-F Next
```

i-0fd3ba4cefceadbdd (minikube)

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Using “kubectl apply -f <filename>” command to create an pods.

```
[ec2-user@ip-172-31-90-195 ~]$ nano Deployment.yaml
[ec2-user@ip-172-31-90-195 ~]$ kubectl apply -f Deployment.yaml
deployment.apps/nginx-deployment created
[ec2-user@ip-172-31-90-195 ~]$ kubectl get pod
NAME                                READY   STATUS    RESTARTS      AGE
mydeployment-5f7f5757b6-2ssgx       1/1     Running   1 (19h ago)   20h
mydeployment-5f7f5757b6-9dwxb       1/1     Running   1 (19h ago)   20h
mydeployment-5f7f5757b6-c6d2d       1/1     Running   1 (19h ago)   20h
mydeployment-5f7f5757b6-kh6sz       1/1     Running   1 (19h ago)   20h
mydeployment-5f7f5757b6-zcfgl       1/1     Running   1 (19h ago)   20h
mynewpod                            1/2     Error     19 (2m ago)   20h
nginx-deployment-59f86b59ff-969ns   1/1     Running   0             26s
nginx-deployment-59f86b59ff-zqbs2   1/1     Running   0             26s
nginx-rs-256c6                      1/1     Running   1 (19h ago)   19h
nginx-rs-dt62w                      1/1     Running   1 (19h ago)   19h
nginx-rs-t4k82                      1/1     Running   1 (19h ago)   19h
[ec2-user@ip-172-31-90-195 ~]$ ls
Deployment.yaml  development.yaml  minikube-linux-amd64  pod.yaml  replicaset.yaml
[ec2-user@ip-172-31-90-195 ~]$ nano service.yaml
```

i-0fd3ba4cefceadbdd (minikube)

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The Service.yaml

```
  GNU nano 8.3                                          service.yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  type: NodePort
  selector:
    app: nginx
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
      nodePort: 30007

                                                  [ Read 13 lines ]
^G Help      ^O Write Out   ^F Where Is    ^K Cut       ^T Execute   ^C Location   M-U Undo      M-A Set Mark   M-] To Bracket  M-B Previous
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify   ^/ Go To Line M-E Redo      M-6 Copy       ^B Where Was    M-F Next
```

i-0fd3ba4cefceadbdd (minikube)

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Using “kubectl apply -f <filename> command to create the service. “kubectl get svc” command to show their services.

```
nginx-rs-t4k82                    1/1     Running   1 (19h ago)      19h
[ec2-user@ip-172-31-90-195 ~]$ kubectl get svc
NAME            TYPE        CLUSTER-IP    EXTERNAL-IP   PORT(S)        AGE
kubernetes      ClusterIP   10.96.0.1     <none>        443/TCP        20h
nginx-service   NodePort    10.96.11.15   <none>        80:30007/TCP   49s
[ec2-user@ip-172-31-90-195 ~]$
```

i-0fd3ba4cefceadbdd (minikube)

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Using “kubectl port-forward svc/nginx-service 8080:80 –address 0.0.0.0” command to using forward portnumber of service in 8080.

```
[ec2-user@ip-172-31-90-195 ~]$ minikube service nginx-service
|-----------|---------------|-------------|---------------------------|
| NAMESPACE |     NAME      | TARGET PORT |            URL            |
|-----------|---------------|-------------|---------------------------|
| default   | nginx-service | 80          | http://192.168.49.2:30007 |
|-----------|---------------|-------------|---------------------------|
* Opening service default/nginx-service in default browser...
  http://192.168.49.2:30007
[ec2-user@ip-172-31-90-195 ~]$ ^C
[ec2-user@ip-172-31-90-195 ~]$ kubectl port-forward svc/nginx-service  8080:80 --address 0.0.0.0
Forwarding from 0.0.0.0:8080 -> 80
^C[ec2-user@ip-172-31-90-195 ~]$ ping
ping: usage error: Destination address required
[ec2-user@ip-172-31-90-195 ~]$ minikube ip
192.168.49.2
[ec2-user@ip-172-31-90-195 ~]$ kubectl port-forward svc/nginx-service  8080:80 --address 0.0.0.0
Forwarding from 0.0.0.0:8080 -> 80
Handling connection for 8080
```

i-0fd3ba4cefceadbdd (minikube)

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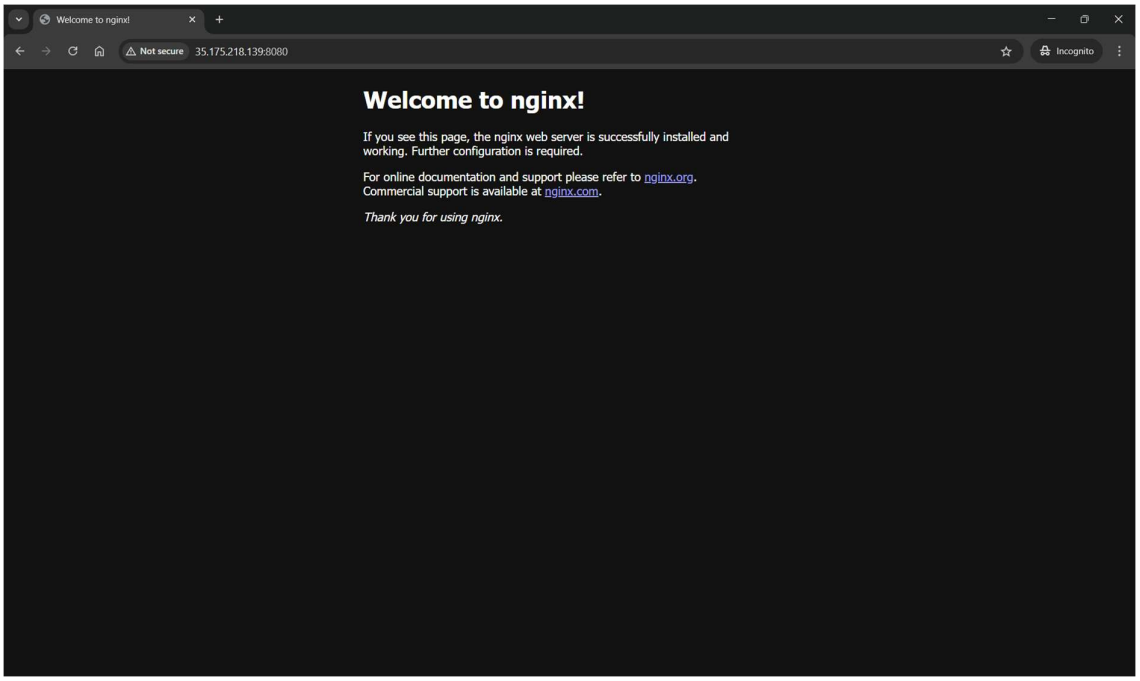
Welcome to nginx!
Not secure
35.175.218.139:8080
Incognito
Welcome to nginx!
If you see this page, the nginx web server is successfully installed and working. Further configuration is required.
For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.
Thank you for using nginx.